

SAFETY DATA SHEET

Sulphur Hexafluoride

Issue Date: 16.01.2013
Last revised date: 31.03.2015

Version: 0.0

SDS No.: 000010021723
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SECTION 1: Identification of the substance / mixture and of the company / undertaking

1.1 Product identifier

Product name: Sulphur hexafluoride

Additional identification

Chemical name: Sulphur hexafluoride

Chemical formula: S F₆

INDEX No. -

CAS-No. 2551-62-4

EC No. 219-854-2

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses: Industrial and professional. Perform risk assessment prior to use.
Insulant.
Use as an Intermediate (transported, on-site isolated).
Use for electronic component manufacture.
Using gas alone or in mixtures for the calibration of analysis equipment.
Using gas for metal treatment.
Formulation of mixtures with gas in pressure receptacles.

Uses advised against Consumer use.

1.3 Details of the supplier of the safety data sheet

Supplier

Linde Malaysia Sdn Bhd (100783-W),
No 13, Jalan 222,
46100 Petaling Jaya,
Selangor Darul Ehsan,
Malaysia

Toll Free: 1800 883 888

E-mail: csc.lg.my@linde.com

1.4 Emergency telephone number:

Emergency phone numbers (24h): 03-79554233

Poison center Unit HAZMAT Malaysia, tel: 999

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 as amended.

Physical Hazards

Gases under pressure

Liquefied gas

H280: Contains gas under pressure; may explode if heated.

2.2 Label Elements



Signal Words: Warning

Hazard Statement(s): H280: Contains gas under pressure; may explode if heated.

Precautionary Statement

Prevention: None.

Response: None.

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Storage: P403: Store in a well-ventilated place.

Disposal: None.

Supplemental label information

EIGA-0783: Contains fluorinated greenhouse gases covered by the Kyoto protocol.
EIGA-As: Asphyxiant in high concentrations.

2.3 Other hazards: Contact with evaporating liquid may cause frostbite or freezing of skin.

SECTION 3: Composition / information on ingredients

3.1 Substances

Chemical name	Sulphur hexafluoride
INDEX No.:	-
CAS-No.:	2551-62-4
EC No.:	219-854-2
Purity:	100%
	The purity of the substance in this section is used for classification only, and does not represent the actual purity of the substance as supplied, for which other documentation should be consulted.
Trade name:	-

SECTION 4: First aid measures

General: In high concentrations may cause asphyxiation. Symptoms may include loss of mobility/consciousness. Victim may not be aware of asphyxiation. Remove victim to uncontaminated area wearing self contained breathing apparatus. Keep victim warm and rested. Call a doctor. Apply artificial respiration if breathing stopped.

4.1 Description of first aid measures

Inhalation:	In high concentrations may cause asphyxiation. Symptoms may include loss of mobility/consciousness. Victim may not be aware of asphyxiation. Remove victim to uncontaminated area wearing self contained breathing apparatus. Keep victim warm and rested. Call a doctor. Apply artificial respiration if breathing stopped.
Eye contact:	Rinse the eye with water immediately. Remove contact lenses, if present and easy to do. Continue rinsing. Flush thoroughly with water for at least 15 minutes. Get immediate medical assistance. If medical assistance is not immediately available, flush an additional 15 minutes.
Skin Contact:	Contact with evaporating liquid may cause frostbite or freezing of skin.
Ingestion:	Ingestion is not considered a potential route of exposure.

4.2 Most important symptoms and effects, both acute and delayed: Respiratory arrest. Contact with liquefied gas can cause damage (frostbite) due to rapid evaporative cooling.

4.3 Indication of any immediate medical attention and special treatment needed

Hazards:	Respiratory arrest. Contact with liquefied gas can cause damage (frostbite) due to rapid evaporative cooling.
Treatment:	Thaw frosted parts with lukewarm water. Do not rub affected area. Get immediate medical advice/attention.

SECTION 5: Firefighting measures

General Fire Hazards: Heat may cause the containers to explode.

5.1 Extinguishing media

Suitable extinguishing media:	Material will not burn. In case of fire in the surroundings: use appropriate extinguishing agent.
Unsuitable extinguishing media:	None.

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5.2 Special hazards arising from the substance or mixture: Fire or excessive heat may produce hazardous decomposition products.

Hazardous Combustion Products: If involved in a fire the following toxic and/or corrosive fumes may be produced by thermal decomposition: hydrogen fluoride ; sulphur dioxide

5.3 Advice for firefighters

Special fire fighting procedures: In case of fire: Stop leak if safe to do so. Continue water spray from protected position until container stays cool. Use extinguishants to contain the fire. Isolate the source of the fire or let it burn out.

Special protective equipment for fire-fighters: Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.
Guideline: EN 469 Protective clothing for firefighters. Performance requirements for protective clothing for firefighting. EN 15090 Footwear for firefighters. EN 659 Protective gloves for firefighters. EN 443 Helmets for fire fighting in buildings and other structures. EN 137 Respiratory protective devices - Self-contained open-circuit compressed air breathing apparatus with full face mask - Requirements, testing, marking.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures: Evacuate area. Provide adequate ventilation. Prevent from entering sewers, basements and workpits, or any place where its accumulation can be dangerous. Wear self-contained breathing apparatus when entering area unless atmosphere is proved to be safe. EN 137 Respiratory protective devices - Self-contained open-circuit compressed air breathing apparatus with full face mask - Requirements, testing, marking.

6.2 Environmental Precautions: Prevent further leakage or spillage if safe to do so.

6.3 Methods and material for containment and cleaning up: Provide adequate ventilation.

6.4 Reference to other sections: Refer to sections 8 and 13.

SECTION 7: Handling and storage:

7.1 Precautions for safe handling: Only experienced and properly instructed persons should handle gases under pressure. Use only properly specified equipment which is suitable for this product, its supply pressure and temperature. Refer to supplier's handling instructions. The substance must be handled in accordance with good industrial hygiene and safety procedures. Protect containers from physical damage; do not drag, roll, slide or drop. Do not remove or deface labels provided by the supplier for the identification of the container contents. When moving containers, even for short distances, use appropriate equipment eg. trolley, hand truck, fork truck etc. Secure cylinders in an upright position at all times, close all valves when not in use. Provide adequate ventilation. Suck back of water into the container must be prevented. Do not allow backfeed into the container. Avoid suckback of water, acid and alkalis. Keep container below 50°C in a well ventilated place. Observe all regulations and local requirements regarding storage of containers. When using do not eat, drink or smoke. Store in accordance with local/regional/national/international regulations. Never use direct flame or electrical heating devices to raise the pressure of a container. Leave valve protection caps in place until the container has been secured against either a wall or bench or placed in a container stand and is ready for use. Damaged valves should be reported immediately to the supplier. Close container valve after each use and when empty, even if still connected to equipment. Never attempt to repair or modify container valves or safety relief devices. Replace valve outlet caps or plugs and container caps where supplied as soon as container is disconnected from equipment. Keep container valve outlets clean and free from contaminants particularly oil and water. If user experiences any difficulty operating container valve discontinue use and contact supplier. Never attempt to transfer gases from one container to another. Container valve guards or caps should be in place.

7.2 Conditions for safe storage, including any incompatibilities: Containers should not be stored in conditions likely to encourage corrosion. Stored containers should be periodically checked for general conditions and leakage. Container valve guards or caps should be in place. Store containers in location free from fire risk and away from sources of heat and ignition. Keep away from combustible material.

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7.3 Specific end use(s): None.

SECTION 8: Exposure controls/ personal protection

8.1 Control Parameters

Occupational Exposure Limits

Chemical name	type	Exposure Limit Values	Source
Sulphur hexafluoride	TWA	2,5 mg / m ³	EU. Indicative Exposure Limit Values in Directives 91/322/EEC, 2000/39/EC, 2006/15/EC, 2009/161/EU (12 2009)

Biological Limit Values

Chemical name	Exposure Limit Values	Source
Sulphur hexafluoride (Fluoride: Sampling time: End of shift.)	7,0 mg / g (Creatinine in urine)	DE BAT (09 2013)
Sulphur hexafluoride (Fluoride: Sampling time: Prior to shift.)	4,0 mg / g (Creatinine in urine)	DE BAT (09 2013)

DNEL-Values

Critical component	type	Value	Remarks
Sulphur hexafluoride	Worker - inhalative, long-term - systemic - Inhalation	77900 mg / m ³	-
	Worker - inhalative, long-term - local - Inhalation	77900 mg / m ³	-

PNEC-Values

Critical component	type	Value	Remarks
Sulphur hexafluoride	Aquatic (intermit. releases)	1,5 mg / l	-
	freshwater	0,15 mg / l	-
	Aquatic (freshwater)	0,15 mg / l	-
	freshwater - intermittent	1,5 mg / l	-

8.2 Exposure controls

Appropriate engineering controls:

Consider a work permit system e.g. for maintenance activities. Ensure adequate air ventilation. Oxygen detectors should be used when asphyxiating gases may be released. Provide adequate ventilation, including appropriate local extraction, to ensure that the defined occupational exposure limit is not exceeded. Systems under pressure should be regularly checked for leakages. Preferably use permanent leak tight connections (eg. welded pipes). Do not eat, drink or smoke when using the product.

Individual protection measures, such as personal protective equipment

General information:

A risk assessment should be conducted and documented in each work area to assess the risks related to the use of the product and to select the PPE that matches the relevant risk. The following recommendations should be considered. Keep self contained breathing apparatus readily available for emergency use. Personal protective equipment for the body should be selected based on the task being performed and the risks involved.

Eye/face protection:

Safety eyewear, goggles or face-shield to EN166 should be used to avoid exposure to liquid splashes. Wear eye protection to EN 166 when using gases.
Guideline: EN 166 Personal Eye Protection.

Skin protection

Hand Protection:

Wear working gloves while handling containers
Guideline: EN 388 Protective gloves against mechanical risks.

Body protection:

No special precautions.

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Other:	Wear safety shoes while handling containers Guideline: EN345 Personal protective equipment - Safety footwear.
Respiratory Protection:	Not required.
Thermal hazards:	No precautionary measures are necessary.
Hygiene measures:	Specific risk management measures are not required beyond good industrial hygiene and safety procedures. Do not eat, drink or smoke when using the product.
Environmental exposure controls:	For waste disposal, see section 13 of the SDS.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance

Physical state:	Gas
Form:	Liquefied gas
Color:	Colorless
Odor:	Odorless
Odor Threshold:	Odor threshold is subjective and is inadequate to warn of over exposure.
pH:	not applicable.
Melting Point:	-50,8 °C
Boiling Point:	-63,8 °C
Sublimation Point:	not applicable.
Critical Temp. (°C):	45,5 °C
Flash Point:	Not applicable to gases and gas mixtures.
Evaporation Rate:	Not applicable to gases and gas mixtures.
Flammability (solid, gas):	Nonflammable Gas
Flammability Limit - Upper (%):	not applicable.
Flammability Limit - Lower (%):	not applicable.
Vapor pressure:	12.038,98 kPa (25 °C)
Vapor density (air=1):	5
Relative density:	1,88 (-50 °C)
Solubility(ies)	
Solubility in Water:	31 mg/l
Partition coefficient (n-octanol / water):	1,68
Autoignition Temperature:	Not applicable.
Decomposition Temperature:	Decomp occurs at high temp in presence of oxygen with release of irritating decomp products. sulfuryl and thionyl fluorides are the major decomp products. When heated to decomp, emits highly toxic fumes of hydrogen fluoride and sulfur oxides.
Viscosity	
Kinematic viscosity:	No data available.
Dynamic viscosity:	0,016 mPa.s (25 °C)
Explosive properties:	Not applicable.
Oxidizing properties:	not applicable.

9.2 Other information:

Molecular weight:	Gas/vapour heavier than air. May accumulate in confined spaces, particularly at or below ground level. 146,06 g/mol (F6S)
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SECTION 10: Stability and reactivity

10.1 Reactivity:	No reactivity hazard other than the effects described in sub-section below.
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10.2 Chemical Stability:	Stable under normal conditions.
10.3 Possibility of Hazardous Reactions:	None.
10.4 Conditions to Avoid:	None.
10.5 Incompatible Materials:	No reaction with any common materials in dry or wet conditions.
10.6 Hazardous Decomposition Products:	Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

General information: None.

11.1 Information on toxicological effects

Acute toxicity - Oral Product	Based on available data, the classification criteria are not met.
Acute toxicity - Dermal Product	Based on available data, the classification criteria are not met.
Acute toxicity - Inhalation Product	Based on available data, the classification criteria are not met.
Repeated dose toxicity Sulphur hexafluoride	NOAEL (Rat(Female, Male), Inhalation): 465.076 mg / m3
Skin Corrosion/Irritation Product	Based on available data, the classification criteria are not met.
Serious Eye Damage/Eye Irritation Product	Based on available data, the classification criteria are not met.
Respiratory or Skin Sensitization Product	Based on available data, the classification criteria are not met.
Germ Cell Mutagenicity Product	Based on available data, the classification criteria are not met.
Carcinogenicity Product	Based on available data, the classification criteria are not met.
Reproductive toxicity Product	Based on available data, the classification criteria are not met.
Specific Target Organ Toxicity - Single Exposure Product	Based on available data, the classification criteria are not met.
Specific Target Organ Toxicity - Repeated Exposure Product	Based on available data, the classification criteria are not met.
Aspiration Hazard Product	Not applicable to gases and gas mixtures.

SECTION 12: Ecological information

12.1 Toxicity

Acute toxicity Product	No ecological damage caused by this product.
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Acute toxicity - Fish

Sulphur hexafluoride LC 50 (Fish, 96 h): 236 mg/l

Acute toxicity - Aquatic Invertebrates

Sulphur hexafluoride LC 50 (Water flea (Daphnia magna), 48 h): 247 mg/l

Toxicity to microorganisms

Sulphur hexafluoride EC50 (Alga, 96 h): 151 mg/l

Additional ecological information

None.

12.2 Persistence and Degradability

Product

Not applicable to gases and gas mixtures..

12.3 Bioaccumulative Potential

Product

The product is expected to biodegrade and is not expected to persist for long periods in an aquatic environment.

12.4 Mobility in Soil

Product

Because of its high volatility, the product is unlikely to cause ground or water pollution.

Sulphur hexafluoride

Henry's Law Constant: 25.347 MPa

12.5 Results of PBT and vPvB assessment

Product

Not classified as PBT or vPvB.

12.6 Other Adverse Effects:

Global Warming Potential

Sulphur hexafluoride

Global warming potential: 22.800

Contains fluorinated greenhouse gases covered by the Kyoto protocol. When discharged in large quantities may contribute to the greenhouse effect. For GWP value of mixture and quantities, refer to container label.

UN / IPCC. Greenhouse Gas Global Warming Potentials (IPCC Fourth Assessment Report. Climate Change, Table TS.2

- Global warming potential: 22800 100-yr

SECTION 13: Disposal considerations

13.1 Waste treatment methods

General information:

Do not discharge into any place where its accumulation could be dangerous. Vent to atmosphere in a well-ventilated place.

Disposal methods:

Refer to the EIGA code of practice (Doc.30 "Disposal of Gases", downloadable at <http://www.eiga.org>) for more guidance on suitable disposal methods. Dispose of container via supplier only. Discharge, treatment, or disposal may be subject to national, state, or local laws.

SECTION 14: Transport information

ADR

14.1 UN Number:	UN 1080
14.2 UN Proper Shipping Name:	SULPHUR HEXAFLUORIDE
14.3 Transport Hazard Class(es)	
Class:	2
Label(s):	2.2
Hazard No. (ADR):	20
Tunnel restriction code:	(C/E)
14.4 Packing Group:	-
14.5 Environmental hazards:	not applicable
14.6 Special precautions for user:	-

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RID

14.1 UN Number: UN 1080
14.2 UN Proper Shipping Name: SULPHUR HEXAFLUORIDE
14.3 Transport Hazard Class(es):
Class: 2
Label(s): 2.2
14.4 Packing Group: -
14.5 Environmental hazards: not applicable
14.6 Special precautions for user: -

IMDG

14.1 UN Number: UN 1080
14.2 UN Proper Shipping Name: SULPHUR HEXAFLUORIDE
14.3 Transport Hazard Class(es):
Class: 2.2
Label(s): 2.2
Ems No.: F-C, S-V
14.3 Packing Group: -
14.5 Environmental hazards: not applicable
14.6 Special precautions for user: -

IATA

14.1 UN Number: UN 1080
14.2 Proper Shipping Name: Sulphur hexafluoride
14.3 Transport Hazard Class(es):
Class: 2.2
Label(s): 2.2
14.4 Packing Group: -
14.5 Environmental hazards: not applicable
14.6 Special precautions for user: -

Other information

Passenger and cargo aircraft: Allowed.

Cargo aircraft only: Allowed.

14.7 Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code: not applicable

Additional identification:

Avoid transport on vehicles where the load space is not separated from the driver's compartment. Ensure vehicle driver is aware of the potential hazards of the load and knows what to do in the event of an accident or an emergency. Before transporting product containers ensure that they are firmly secured. Ensure that the container valve is closed and not leaking. Container valve guards or caps should be in place. Ensure adequate air ventilation.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture.:

National Regulations

Occupational Safety and Health Act 1994 and relevant regulations:
Occupational Safety and Health (Classification, Labeling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.
Occupational Safety and Health (Use And Standards of Exposure of Chemicals Hazardous to Health) Regulations 2000
Environment Quality Act 1974 & regulations:
Environment Quality (Clean Air) Regulations 2014
Environmental Quality (Scheduled Wastes) Regulations 2005

15.2 Chemical safety assessment:

CSA has been carried out.

SECTION 16: Other information

Training information:

Users of breathing apparatus must be trained. The hazard of asphyxiation is often overlooked and must be stressed during operator training. Ensure operators understand the hazards.

Other information:

Before using this product in any new process or experiment, a thorough material compatibility and safety study should be carried out. Ensure adequate air ventilation. Ensure all national/local regulations are observed. Whilst proper care has been taken in the preparation of this document, no liability for injury or damage resulting from its use can be accepted.

Disclaimer:

This information is provided without warranty. The information is believed to be correct. This information should be used to make an independent determination of the methods to safeguard workers and the environment.